

Nhlanzeko Lwazi Mhlongo

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TECHNICAL SKILLS

Languages: Java (21), TypeScript, JavaScript (ES6+), PHP, Python, SQL, HTML5, CSS3.

Frameworks & Libraries: Spring Boot 3, Next.js 14, React, Tailwind CSS.

Databases: PostgreSQL (Supabase, Neon), MySQL, MongoDB, Firebase, Hibernate/JPA.

Tools & Platforms: Docker, Git, GitHub, GitHub Actions, Vercel, Maven, IntelliJ IDEA, Google Cloud AI.

Methodologies: Agile/Scrum, CI/CD

EDUCATION

Bachelor of Science in Information Technology - [Eduvos]

Feb 2023 – May 2026 | Final Average: 3.58 GPA (76.3%)

- Relevant Modules: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Architecture, Web Development, Operating Systems
- Achievements: 18 Module Distinctions

PROJECTS

Personal Portfolio Website | (www.lwazimhlongo.com / [Github](#))

Built with: React.js, Tailwind CSS, Vercel

- Engineered a high-performance personal portfolio using React architecture and Tailwind CSS for responsive design.
- Configured a CI/CD pipeline using GitHub Actions to automate build and deployment workflows, ensuring seamless updates.
- Implemented advanced UX features including a lanyard, slide-up/slide-down page transitions and smooth scrolling navigation.
- Developed a serverless contact form backend to replace legacy email links, improving user engagement and security.
- Optimized SEO and meta tags to ensure professional indexing and social sharing preview performance.
- Managed full-stack deployment configuration, including custom domain DNS settings and HTTPS security certification.
- Connected Google Analytics to monitor user behaviour and traffic.

Atomic Ledger | AI-Powered Banking Switch ([Demo](#) / [GitHub](#))

Built with: Java 21, Spring Boot 3, Google Gemini AI, PostgreSQL (Supabase), Docker

- Engineered a high-performance financial switching engine capable of routing transactions between multiple banking ledgers (Investec, Absa, Standard Bank) based on BIN validation logic.
- Integrated Google Gemini 2.0 Flash to autonomously categorize transaction metadata in real-time, implementing resilient error handling to ensure 99.99% system uptime.
- Optimized API latency by implementing Spring Caching (In-Memory), reducing redundant AI calls and accelerating response times for recurring transaction types.
- Secured sensitive credentials using environment-based configuration (application.properties) and implemented strict input validation to prevent negative-balance fraud.
- Deployed a cloud-native architecture using Docker containers on Render, connected to a managed Supabase PostgreSQL cluster for immutable transaction logging.

Humble Coffee | E-commerce Platform ([Demo](#) / [GitHub](#))

Built with: Next.js, TypeScript, Tailwind CSS, PostgreSQL (Supabase)

- Developed a full-stack e-commerce solution featuring a dynamic pricing engine to handle complex product variants (e.g., weighted items, gift cards).
- Engineered advanced cart logic with LocalStorage persistence to ensure a seamless user experience across sessions.
- Architected a scalable database schema on Supabase (PostgreSQL) to manage real-time inventory and product data.
- Implemented secure user authentication and row-level security (RLS) to protect sensitive customer data and transaction history.
- Optimized mobile performance using adaptive media loading techniques to improve core web vitals.

StudyConnect | Student Networking PWA ([Demo](#) / [GitHub](#))

Built with: Next.js, TypeScript, Tailwind CSS

- Migrated legacy native Android architecture to a high-performance Progressive Web App (PWA) using Next.js 14 to eliminate installation barriers.
- Leveraged TypeScript for strict type safety, ensuring robust code maintainability and reducing runtime errors during scaling.
- Implemented a "Find Peers" discovery engine allowing users to filter connections by university, module, or study interests.
- Engineered a custom "Digital Phone" UI chassis with Glassmorphism aesthetics (backdrop-blur) to simulate native mobile interactions within a web browser.
- Implemented independent viewport scrolling and touch-simulation logic to replicate the "feel" of a native mobile app on desktop interfaces.

FinTech Loan Scorer - Credit Risk Assessment Engine ([Demo](#) / [GitHub](#))

Tech Stack: Python, FastAPI, React, SQLite

- Engineered a full-stack credit risk assessment platform, developing a custom Python-based weighted scoring algorithm to process and evaluate complex financial data.
- Implemented programmatic stress testing methodologies to validate the resilience and accuracy of the algorithmic logic under varied data conditions.
- Designed the backend architecture with FastAPI and SQLite to ensure efficient data manipulation, persistent audit trails, and seamless integration with the user-facing dashboard.

Asset Compass - Full-Stack FinTech Portfolio Tracker ([Demo](#) / [GitHub](#))

Tech Stack: Next.js, TypeScript, Tailwind CSS, Java 21, Spring Boot 3, PostgreSQL

- Engineered a full-stack financial dashboard utilizing the AlphaVantage API to fetch real-time market data and foreign exchange rates (USD/ZAR).
- Architected a "Circuit Breaker" fallback mechanism to catch third-party API rate limits, seamlessly serving deterministic mock data to ensure zero application downtime.
- Implemented secure, stateless JWT authentication via Spring Security with BCrypt password hashing for an optimized user onboarding flow.
- Developed an automated event-driven audit trail that logs asset transactions and price refreshes, rendering dynamic performance graphs via Recharts.
- Optimized serverless database stability by configuring custom HikariCP connection pooling, proactively preventing idle-timeout connection drops on Neon PostgreSQL.

CERTIFICATIONS & CONTINUOUS LEARNING

- **HackerRank** – Java (Basics) (2026)

LEADERSHIP & ACTIVITIES

Eduvos Chess Club Leader: 2024 – 2025

- Led the university club to a 2nd place victory in total wins at the Rosebank College Tournament, fostering team collaboration and strategic problem-solving.